

1. (20 marks) For  $(a, b) \in \mathbb{R}^2$ , consider the following quadratic form

$$Q(\mathbf{x}) = b(x_1^2 + x_2^2 + x_3^2) + 2a(x_1x_2 + x_2x_3).$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients*  $A$  associated with this quadratic form. That is, find the matrix  $A$  satisfying  $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ . Then, find all the principal minors of  $A$ .

**Solutions:** (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} b & a & 0 \\ a & b & a \\ 0 & a & b \end{pmatrix}$$

(2 marks) Its third order principal minor equals  $b^3 - 2ba^2$ . Its second order principal minors are equal to  $b^2 - a^2$  and  $b^2$ . Its first order principal minors are equal to  $b$ .

- (b) (12 marks) Define what it means for  $A$  to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all parameter values for  $a$  and  $b$  such that the quadratic form  $Q(\mathbf{x})$  is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*?

**Solutions:** (2 marks) A quadratic form is indefinite if  $Q(\mathbf{x}) > 0$  and  $Q(\mathbf{y}) < 0$  for some  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n \setminus \mathbf{0}$ . A quadratic form is positive semidefinite if  $Q(\mathbf{x}) \geq 0$  for all  $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$ . A quadratic form is negative semidefinite if  $Q(\mathbf{x}) \leq 0$  for all  $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$ .

(4 marks)  $Q(\mathbf{x})$  is positive semidefinite if all principal minors are positive which requires:

$$b \geq 0 \quad \text{and} \quad a \in \left[-b/\sqrt{2}, b/\sqrt{2}\right].$$

The condition accounts for scenarios in which some of the parameters are equal to 0.

(4 marks)  $Q(\mathbf{x})$  is negative semidefinite if all principal minors suitably alternate in sign which requires:

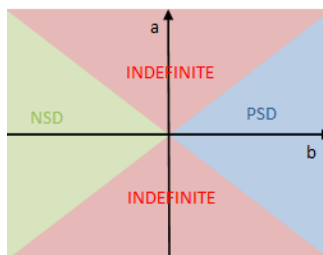
$$b \leq 0 \quad \text{and} \quad a \in \left[b/\sqrt{2}, -b/\sqrt{2}\right].$$

The condition accounts for scenarios in which some of the parameters are equal to 0.

(2 marks)  $Q(\mathbf{x})$  is indefinite whenever it is not semidefinite.

- (c) (4 marks) Plot the pairs  $(a, b)$  for which  $Q(\mathbf{x})$  is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in  $\mathbb{R}^2$ .

**Solutions:** (4 marks) The plot below depicts the answer



2. (20 marks) Consider the functions  $f(\mathbf{x}) = -x_1^2 - x_2^2$  defined on some convex set  $U$  in  $\mathbb{R}^2$ .

- (a) (6 marks) Define what it means for  $f$  to be *quasi-concave* and *quasi-convex*.

**Solutions:** The function  $f$  is: (3 marks) *quasi-concave* on  $U$  if  $C_k^+ = \{\mathbf{x} \in U \mid f(\mathbf{x}) \geq k\}$  is a convex set for every real number  $k$ ; (3 marks) *quasi-convex* on  $U$  if  $C_k^- = \{\mathbf{x} \in U \mid f(\mathbf{x}) \leq k\}$  is a convex set for every real number  $k$ .

- (b) (14 marks) Is  $f$  *quasi-concave* or *quasi-convex*? Hint: Draw level sets to gain intuition and use counterexamples to show that a property fails for some  $U$ .

**Solutions:** (9 marks) The function  $f$  is quasi-concave. To show this consider  $\mathbf{x}, \mathbf{y} \in C_k^+$  for some  $k \in \mathbb{R}_+$  (upper-countour sets are empty and thus convex for  $k > 0$ ). Let  $\mathbf{z} = a\mathbf{x} + (1-a)\mathbf{y}$  for some  $a \in [0, 1]$ . If so, we have that

$$\begin{aligned} f(\mathbf{z}) &= -(ax_1 + (1-a)y_1)^2 - (ax_2 + (1-a)y_2)^2 = \\ &= a^2 f(\mathbf{x}) + (1-a)^2 f(\mathbf{y}) - a(1-a)(2x_1y_1 + 2x_2y_2) \\ &\geq a^2 f(\mathbf{x}) + (1-a)^2 f(\mathbf{y}) - a(1-a)(x_1^2 + y_1^2 + x_2^2 + y_2^2) \\ &= a^2 f(\mathbf{x}) + (1-a)^2 f(\mathbf{y}) + a(1-a)(f(\mathbf{x}) + f(\mathbf{y})) \\ &\geq a^2 k + (1-a)^2 k + a(1-a)(k + k) = k, \end{aligned}$$

where the second inequality follows from  $\mathbf{x}, \mathbf{y} \in C_k^+$ , while the first one follows from

$$(x_i - y_i)^2 \geq 0 \text{ which implies that } x_i^2 + y_i^2 \geq 2x_iy_i.$$

(5 marks) The function  $f$  is not quasi-convex. To show this consider  $\mathbf{x} = (1, 1)$  and  $\mathbf{y} = (-1, -1)$  and let  $\mathbf{z} = \mathbf{x}/2 + \mathbf{y}/2 = (0, 0) \in \mathbb{R}^2$ . If so, we have that

$$\mathbf{x}, \mathbf{y} \in C_{-2}^- \text{ and yet } \mathbf{z} \notin C_{-2}^-,$$

since  $f(\mathbf{z}) = 0$ .

3. (40 marks) For  $a > 0$ , consider the problem:

$$\max_{x,y} \sqrt{xy} \text{ s.t. } ax + y \leq 16, \quad x \geq a, \quad y \geq 0.$$

- (a) (10 marks) Is the objective function concave? Are the constraints convex? For what values of  $a > 0$  is Slater condition satisfied? So, for what values of  $a > 0$ , can you solve this optimization problem using the Kuhn-Tucker method?

**Solutions:** (5 marks) The objective function is concave since

$$\begin{aligned} D^2 f(x, y) &= \begin{pmatrix} -x^{-3/2}y^{1/2}/4 & x^{-1/2}y^{-1/2}/4 \\ x^{-1/2}y^{-1/2}/4 & -y^{-3/2}x^{1/2}/4 \end{pmatrix}. \\ |D^2 g_1(x, y)| &= 0, \quad -x^{-3/2}y^{1/2}/4 \leq 0, \quad -y^{-3/2}x^{1/2}/4 \leq 0. \end{aligned}$$

The constraints are linear and thus convex.

(5 marks) Thus, when the constraint set has a non-empty interior, we can adopt the Kuhn-Tucker approach to solve the problem. This is the case for all  $a \in (0, 4)$ , since the point  $\tilde{x} = a + \varepsilon$  and  $\tilde{y} = \varepsilon$  always belongs to the interior of the constraint set for  $\varepsilon$  sufficiently small as long as  $a < 4$ , since

$$a\tilde{x} + \tilde{y} = a^2 + \varepsilon(a+1) < 16, \quad \tilde{x} > a, \quad \tilde{y} > 0.$$

- (b) (6 marks) Suppose the restrictions on  $a$  found in part (a) hold. Write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?

**Solutions:** (5 marks) The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= x^{-1/2}y^{1/2}/2 - a\lambda_1 + \lambda_2, \\ 0 &= x^{1/2}y^{-1/2}/2 - \lambda_1 + \lambda_3, \\ 0 &= \lambda_1(ax + y - 16) = \lambda_2(a - x) = \lambda_3y, \\ 0 &\geq ax + y - 16, 0 \geq a - x, 0 \geq -y, \\ 0 &\leq \lambda_1, 0 \leq \lambda_2, 0 \leq \lambda_3. \end{aligned}$$

(1 marks) Any solutions to these conditions is maximizer of the Lagrangian, because the Lagrangian is a sum of concave functions by the arguments in (a) and because Slater condition holds.

- (c) (18 marks) Suppose the restrictions on  $a$  found in part (a) hold. Solve the constrained optimization problem in terms of  $a$ .

**Solutions:** Consider the various supports:

- (3 marks) At the solution,  $\lambda_3(a) = 0$ . Otherwise,  $f(x(a), y(a)) = 0$  which cannot be optimal, since the point  $\tilde{x} = a + \varepsilon$  and  $\tilde{y} = \varepsilon$  always belongs to the constraint set for  $\varepsilon$  sufficiently small and

$$f(\tilde{x}, \tilde{y}) > 0.$$

- (3 marks) At the solution,  $\lambda_1(a) > 0$ . To show this, posit that  $\lambda_1(a) = 0$ , and consider any hypothetical solution  $(x(a), y(a))$  and the point  $\tilde{x} = x(a)$  and  $\tilde{y} = 16 - ax(a) > y(a)$ . Such point always belongs to the constraint set and

$$f(\tilde{x}, \tilde{y}) > f(x(a), y(a)),$$

leading to a contradiction.

- (6 marks) If  $\lambda_3(a) = 0$  and  $\lambda_1(a), \lambda_2(a) > 0$ , then by complementary slackness  $x(a) = a$  and  $y(a) = 16 - a^2$  for  $a < 4$ . Further, FOC would then imply that

$$\begin{aligned} \lambda_1(a) &= \frac{a^{1/2}}{2(16 - a^2)^{1/2}} > 0 \\ \lambda_2(a) &= \frac{a^{3/2}}{2(16 - a^2)^{1/2}} - \frac{(16 - a^2)^{1/2}}{2a^{1/2}} \geq 0 \Leftrightarrow 2a^2 \geq 16. \end{aligned}$$

So as multipliers are non-negative, this is a solution for  $a \in [2\sqrt{2}, 4)$ .

- (6 marks) Finally, if  $\lambda_3(a) = \lambda_2(a) = 0$  and  $\lambda_1(a) > 0$ , then FOC and the constraints would imply that

$$\begin{aligned} y(a) &= 8 \geq 0, \\ x(a) &= 8/a \geq a \Leftrightarrow a \leq 2\sqrt{2}, \\ \lambda_1(a) &= \frac{1}{2\sqrt{a}} \geq 0. \end{aligned}$$

So as constraints need to be satisfied, this is a solution for  $a \in (0, 2\sqrt{2}]$ .

- (d) (6 marks) Suppose the restrictions on  $a$  found in part (a) hold. Use the Envelope Theorem to express the slope of value function with respect to the parameter  $a$  in terms of the solution  $(x(a), y(a), \boldsymbol{\lambda}(a))$  found in part (c).

**Solutions:** (6 marks) The Envelope Theorem states that if  $(x(a), y(a), \boldsymbol{\lambda}(a))$  are differentiable functions and the constraint qualification holds, then

$$\frac{dv(a)}{da} = \left. \frac{\partial L(x, y, \boldsymbol{\lambda}|a)}{\partial a} \right|_{(x, y, \boldsymbol{\lambda})=(x(a), y(a), \boldsymbol{\lambda}(a))} = -\lambda_1(a)x(a) - \lambda_2(a) < 0.$$

4. Consider the correspondence  $\mathbf{h} : [0, 4]^2 \rightrightarrows [0, 4]^2$ , where  $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$  and

$$h_1(\mathbf{x}) = \begin{cases} [0, 2] & \text{if } x_2 = 0 \\ [1, 2] & \text{if } x_2 \in (0, 4) \\ [1, 4] & \text{if } x_2 = 4 \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} [0, 3] & \text{if } x_1 = 0 \\ [2, 3] & \text{if } x_1 \in (0, 4) \\ [2, 4] & \text{if } x_1 = 4 \end{cases}.$$

Argue why *Kakutani fixed point theorem* applies. Then, find all the *fixed points* of this correspondence.

**Solutions:** (4 marks) Both correspondences have closed graph and are therefore UHC. Moreover, both are obviously convex-valued and non-empty. Since the image of  $\mathbf{h}$  belongs to  $[0, 4]^2$  which is also the domain, the Fixed Point Theorem applies. Thus, there will be at least one fixed point on  $[0, 4]^2$ .

(3 marks) First, consider  $(0, 0)$ . Such a point is a fixed point of  $\mathbf{h}$  since  $(0, 0) \in \mathbf{h}(0, 0)$ . (1 mark) Further, there can be no other fixed point when  $x_2 = 0$  since  $h_1(x_1, 0) = [0, 2]$  and  $0 \notin h_2(x_1, x_2)$  for  $x_1 > 0$ .

(3 marks) Now, consider  $(4, 4)$ . Such a point is a fixed point of  $\mathbf{h}$  since  $(4, 4) \in \mathbf{h}(4, 4)$ . (1 mark) Further, there can be no other fixed point when  $x_2 = 4$  since  $h_1(x_1, 4) = [1, 4]$  and  $4 \notin h_2(x_1, x_2)$  for  $x_1 < 4$ .

(6 marks) Finally, consider  $\mathbf{x} \in [1, 2] \times [2, 3]$ . Such points are fixed points of  $\mathbf{h}$  since

$$\begin{aligned} [1, 2] &= h_1(\mathbf{x}) \quad \text{for } x_2 \in [2, 3], \\ [2, 3] &= h_2(\mathbf{x}) \quad \text{for } x_1 \in [1, 2]. \end{aligned}$$

(1 mark) Further, there can be no other fixed point when  $x_2 \in [2, 3]$  since  $h_1(x_1, x_2) = [1, 2]$ .

(1 marks) There are no fixed points in which  $x_2 \in (1, 2) \cup (3, 4)$  since  $h_1(\mathbf{x}) = [1, 2]$  and  $h_2(x_1, x_2) = [2, 3]$  for any  $x_1 \in [1, 2]$ .